

Quantum-Well Comb DWDM Transmission via Residual Carrier-Based Phase Recovery and EEPN Mitigation

Xiang Cai^(1,5), Yueying Niu^(2,5), Yixiao Zhu^(3*), Tianchi Zhong⁽¹⁾, Defan Sun⁽²⁾, Tianhong Zhang⁽¹⁾, Fanghuasong Lin⁽²⁾, Hao Song⁽²⁾, Xiansong Fang⁽¹⁾, Xian Zhou⁽⁴⁾, Dan Lu^(2*), and Fan Zhang^(1*)

⁽¹⁾ State Key Laboratory of Photonics and Communications, School of Electronics, Peking University, Beijing 100871, China, *fzhang@pku.edu.cn

⁽²⁾ State Key Laboratory of Optoelectronic Materials and Devices, Institute of Semiconductors, Chinese Academy of Sciences, Beijing 100083, China, *ludan@semi.ac.cn

⁽³⁾ State Key Laboratory of Photonics and Communications, Department of Electronic Engineering, Shanghai Jiao Tong University, Shanghai 200240, China, *yixiaozhu@sjtu.edu.cn

⁽⁴⁾ University of Science and Technology Beijing, Beijing 100083, China

⁽⁵⁾ These authors contributed equally

Abstract We experimentally demonstrate $8\lambda \times 70$ -GBd 16QAM DWDM transmission over 1280-km SSF using quantum-well mode-locked laser combs. We employ residual comb line-based phase tracking to address the 9.0-MHz and 54.6-MHz comb linewidths, achieving >6-dB Q²-factor improvement at BTB and 3.6-dB additional gain from EEPN after 1280-km transmission. ©2026 The Author(s)

Introduction

The increasing demand for transmission capacity in wavelength-division multiplexed (WDM) systems drives the need for efficient multi-wavelength optical sources. As the WDM channel count increases, conventional laser arrays suffer from rapidly growing complexity and energy consumption, whereas a single optical frequency comb (OFC) source can provide multiple carriers with precise frequency spacing, enabling dense WDM (DWDM) with high spectral efficiency [1,2].

Quantum-well semiconductor mode-locked lasers (QW-SMLLs) generate highly coherent OFCs with wide spectral coverage and high output power. Their repetition rates can be flexibly tuned from 10-GHz to 200-GHz, offering a large number of usable comb lines, which makes them highly attractive for coherent optical transmission systems [3,4]. As promising OFC sources, QW-SMLLs exhibit compact configuration, cost-effectiveness, and excellent stability [5,6]. However, traditional coherent systems require narrow laser linewidths. Although adjacent QW comb lines exhibit low differential phase noise, the absolute phase noise of individual lines remains high, with tens of MHz linewidths. In coherent systems, such high phase noise induces random constellation rotation and severe equalization-enhanced phase noise (EEPN) in long-haul links [7,8].

In this work, we experimentally demonstrate, to the best of our knowledge, the first long-haul coherent transmission using QW-SMLL combs. We previously proposed a residual carrier-based phase recovery and validated it in short-reach links [9]. We now extend this to long-haul WDM transmission to address 9.0-MHz and 54.6-MHz QW comb linewidths under dispersion and fiber nonlinear phase noise. The residual optical comb lines serve as a phase reference, enabling

effective frequency-phase recovery without additional time-domain or frequency-domain pilots. Residual carrier recovery can remove phase noise before dispersion compensation, thereby mitigating EEPN induced by the ordering mismatch of local oscillator (LO) phase noise and dispersion, and significantly reducing the receiver LO linewidth requirement [10]. For $8\lambda \times 70$ -GBd 16QAM DWDM system using QW combs, our method achieves >6-dB Q²-factor improvement at back-to-back (BTB) and 3.6-dB additional gain from EEPN mitigation after 1280-km standard single-mode fiber (SSMF) transmission.

Principle of EEPN mitigation

Fig. 1(a) illustrates the interplay between phase noise and chromatic dispersion in conventional coherent systems. Phase recovery is typically based on known symbols embedded in the recovered signal as pilots, and therefore needs to be performed after dispersion compensation. The recovered signal $R_1(t)$ can be expressed as

$$R_1(t) = S(t) \cdot e^{j\phi_{\text{Tx}}(t)} * h_{\text{CD}}(t) \cdot e^{-j\phi_{\text{Tx}}(t)} * h_{\text{CD}}^{-1}(t) \cdot e^{-j[\phi_{\text{Tx}}(t) - \phi_{\text{Tx}}(t)]}, (1)$$

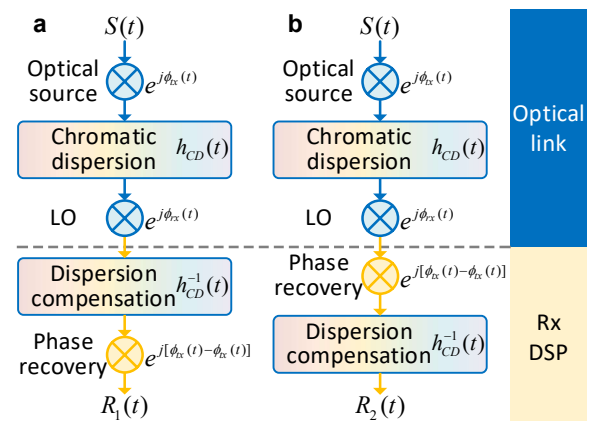


Fig. 1: Illustration of phase noise and chromatic dispersion interplay in (a) conventional and (b) residual carrier-based coherent systems. LO: local oscillator; Rx: receiver-side.

where $S(t)$ is the transmitted signal, $h_{CD}(t)$ is the impulse response of chromatic dispersion, $\phi_{tx}(t)$ and $\phi_{rx}(t)$ denote the phase noise of the transmitter optical source and the receiver LO, respectively. In Eq. (1), the phase noise multiplication and dispersion convolution are non-commutative. As a result, the phase noise spreads over the signal after dispersion compensation, manifesting as EEPN with noise variance derived as [7,8]

$$\sigma_{EEPN}^2 = \pi^2 \cdot |\beta_2| \cdot L \cdot R_s \cdot LW, \quad (2)$$

where β_2 denotes the group-velocity dispersion parameter, L is the transmission distance, R_s is the symbol rate, and LW is the LO laser linewidth. Therefore, EEPN becomes more pronounced in high-symbol-rate and long-haul systems, especially with a broader LO linewidth.

Fig. 1(b) illustrates the interplay between phase noise and chromatic dispersion in the proposed EEPN mitigation scheme. The residual carrier-based recovery does not rely on embedded pilots [9] and enables phase noise extraction and removal before dispersion compensation. As a result, the ordering mismatch between LO phase noise and dispersion is avoided, thereby mitigating EEPN. The corresponding recovered signal $R_2(t)$ can be expressed as

$$R_2(t) = S(t) \cdot e^{j\phi_{tx}(t)} \cdot h_{CD}(t) \cdot e^{-j\phi_{rx}(t)} \cdot e^{-j[\phi_{tx}(t) - \phi_{rx}(t)]} \cdot h_{CD}^{-1}(t) \quad (3)$$

$$= S(t) \cdot e^{j\phi_{tx}(t)} \cdot h_{CD}(t) \cdot e^{-j\phi_{rx}(t)} \cdot h_{CD}^{-1}(t)$$

From Eq. (3), the recovered signal $R_2(t)$ is decoupled from the receiver LO phase noise $\phi_{rx}(t)$, but remains affected by the transmitter phase noise $\phi_{tx}(t)$. In typical long-haul WDM systems, the receiver LO linewidth is more cost-sensitive. According to Eq. (2), when $LW_{tx} < LW_{rx}$, EEPN can be effectively mitigated.

Experimental Setup, Device Characterization, and DSP of Residual Carrier-Based Recovery

The experimental setup is shown in Fig. 2(a). At the transmitter, a QW comb is used as the multi-wavelength optical source. The comb output is amplified by a polarization-maintaining erbium-doped fiber amplifier (PM-EDFA) and filtered by a wavelength-selective switch (WSS, Finisar 4000s) for comb-line selection and power flattening. The baseband signals are generated offline by the transmitter-side (Tx) digital signal processing (DSP) shown in Fig. 2(e), uploaded to a 4-channel 120-GSa/s arbitrary waveform generator (AWG, Keysight M8194), amplified by 50-GHz bandwidth electrical amplifiers (EA), and applied to ~ 27 -GHz bandwidth IQ modulators to generate $8\lambda \times 70$ -GBd 16QAM signals, with odd- and even-channel interleaving. Residual comb lines are introduced by slightly offsetting the bias point of modulators from the null point. Polarization-division-multiplexing (PDM) is emulated using a polarization beam combiner and a ~ 10 -ns decorrelation delay line. The transmission link is a re-circulating loop consisting of 4 spans of 80-km SSMF, with EDFAs placed after each span to compensate for ~ 15 -dB loss. At the receiver, another QW comb is used as the multi-wavelength LO. The received signal is filtered by an optical band-pass filter (OBPF) to extract the channel under test, detected by a standard coherent receiver with 70-GHz balanced photodiodes (BPD), and sampled by a 4-channel 256-GSa/s real-time oscilloscope (RTO, Keysight UXR0594AP) for receiver-side (Rx) offline DSP shown in Fig. 2(f).

The monolithically integrated QW-SMLLs are self-developed packaged devices with identical

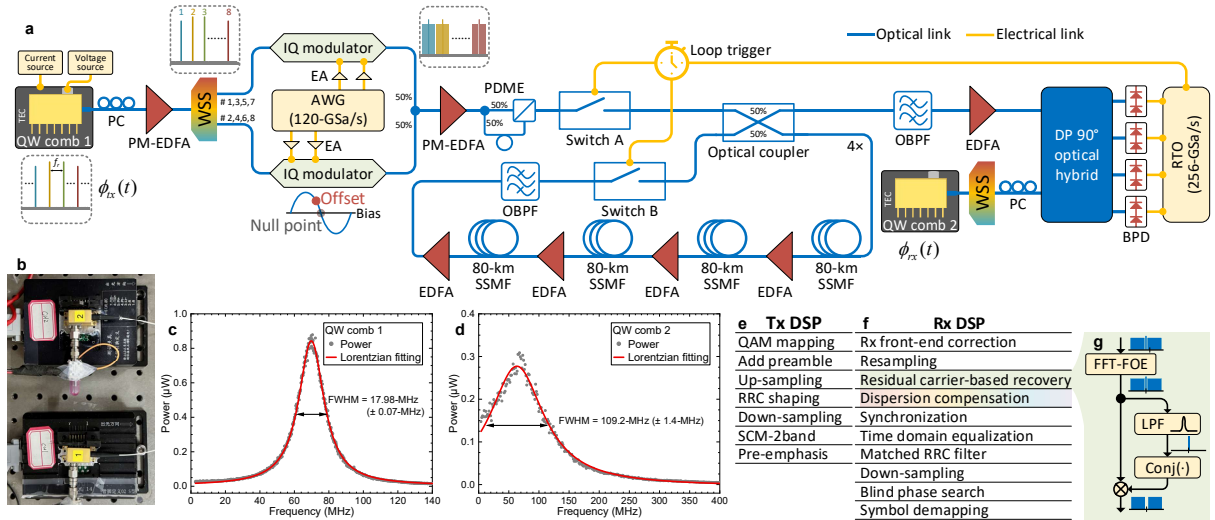


Fig. 2: (a) Experimental setup. QW: quantum-well; TEC: temperature controller; PC: polarization controller; PM: polarization-maintaining; EDFA: erbium-doped fiber amplifier; WSS: wavelength-selective switch; AWG: arbitrary waveform generator; PDME: polarization-division-multiplexing emulator; SSMF: standard single-mode fiber; OBPF: optical band-pass filter; BPD: balanced photodiode; RTO: real-time oscilloscope. (b) Photograph of the packaged QW semiconductor mode-locked laser combs. Delayed self-heterodyne interferometer power spectra and Lorentzian fittings for (c) QW comb 1 and (d) QW comb 2. FWHM: full width at half maximum. (e) Transmitter-side and (f) Receiver-side DSP stacks. RRC: root-raised cosine; SCM: subcarrier multiplexing. (g) Principle of residual carrier-based frequency offset removal and phase recovery. FFT-FOE: FFT peak-based frequency offset estimation; Conj: conjugate.

configurations, as shown in Fig. 2(b). They adopt AlGaInAs material and feature a total cavity length of 1750- μm , corresponding to a repetition rate of 25.6-GHz. After WSS filtering, stable 77-GHz-spaced OFCs are obtained, suitable as multi-wavelength sources for the ~ 76 -GHz bandwidth-per-channel DWDM system (70-GBd, 5-GHz guard band for residual carrier, 0.01 RRC roll-off). The measured delayed self-heterodyne interferometer (DSHI) power spectra and corresponding Lorentzian fittings for QW combs are shown in Fig. 2(c) and Fig. 2(d), yielding 9.0-MHz linewidth of the transmitter optical source and 54.6-MHz linewidth of the receiver LO.

Fig. 2(g) illustrates the principle of residual carrier-based recovery. The residual comb lines generate distinct spectral peaks for frequency offset estimation. After down-conversion, a digital low-pass filter (LPF) extracts the residual carrier, which is then beat with the signal to remove residual frequency offset and phase noise before dispersion compensation.

Results and Discussion

Fig. 3(a) shows the measured optical spectra of the WDM signals at different transmission distances, with a resolution of 0.02-nm. Fig. 3(b) shows that the optimal launch power is 1-dBm per channel after 960-km and 1280-km SSMF transmission. The proposed EEPN mitigation scheme provides consistent gain across the launch power range, in contrast to nonlinearity. Fig. 3(c) shows the measured BER of all the 8 channels over 960-km SSMF transmission, labeled from short to long wavelength. All channels achieve the 20% SD-FEC threshold.

Fig. 3(d) shows the measured Q^2 -factor of the conventional time-domain pilot-based recovery versus pilot ratio at BTB. All points are obtained under the fixed line rate of 560-Gb/s, such that

increasing the pilot ratio reduces the net data rate. Despite this trade-off, the pilot-based recovery fails to compensate phase noise with 63.6-MHz total linewidth. For comparison, the proposed residual carrier-based recovery is also included in Fig. 3(d), achieving >6 -dB Q^2 -factor improvement without relying on additional pilots. With the optimized launch power, the measured Q^2 -factor versus transmission distance is shown in Fig. 3(e). Residual carrier-based recovery enables 960-km transmission with 20% SD-FEC threshold and 1280-km transmission with 24% SD-FEC threshold [11], achieving 3.6-dB additional Q^2 -factor gain from EEPN mitigation.

The recovered constellations of Channel-5 at BTB are shown in Fig. 3(f) for pilot-based recovery and residual carrier (RC)-based recovery. The constellations after 960-km and 1280-km SSMF transmission, with (w/) and without (w/o) EEPN mitigation, are shown in Fig. 3(g) and Fig. 3(h), respectively.

Conclusions

We experimentally demonstrate, to the best of our knowledge, the first long-haul coherent transmission using QW-SMLL combs. We leverage residual comb line-based phase tracking for joint frequency-phase recovery and EEPN mitigation. For $8\lambda \times 70$ -GBd 16QAM DWDM transmission using QW combs with 9.0-MHz and 54.6-MHz linewidths, the residual carrier-based recovery achieves >6 -dB Q^2 -factor improvement at BTB and 3.6-dB additional gain from EEPN mitigation after 1280-km SSMF transmission. Phase recovery before dispersion compensation decouples LO phase noise from dispersion. These results mitigate the phase-noise limitation of QW lasers and highlight their potential as scalable and cost-effective multi-wavelength sources for future WDM coherent transmission.

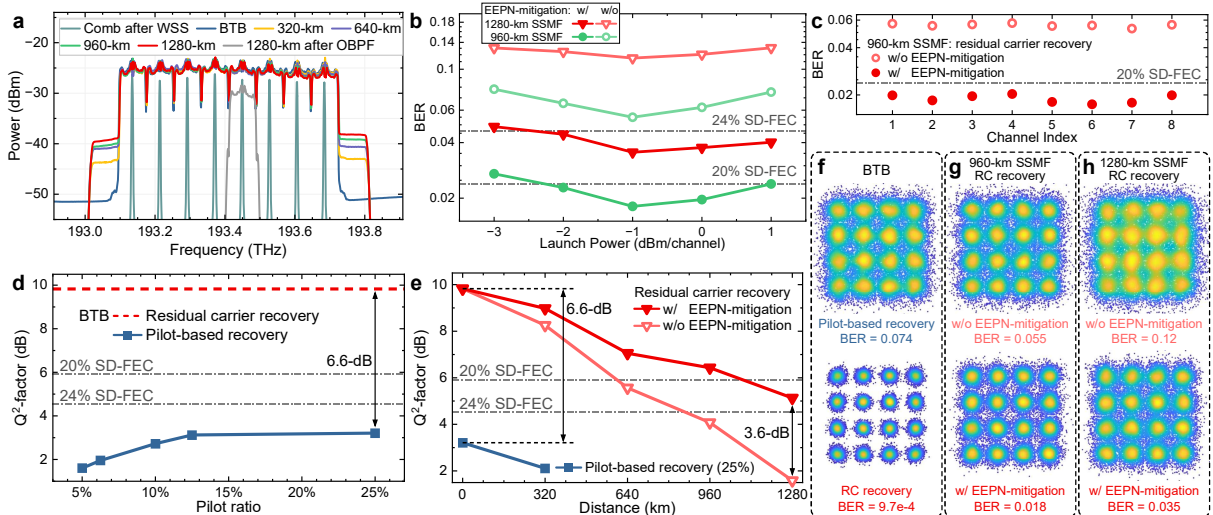


Fig. 3: (a) Measured optical spectra of WDM signals at different transmission distances. (b) Measured BER versus launch power per channel over 960-km and 1280-km SSMF transmission. (c) Measured BER versus channel index. (d) Measured Q^2 -factor versus time-domain pilot ratio of the pilot-based recovery at BTB, with residual carrier-based recovery shown for comparison. (e) Measured Q^2 -factor versus transmission distance. Recovered constellations of Channel-5 (f) at BTB, after (g) 960-km and (h) 1280-km SSMF transmission. RC: residual carrier.

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References

- [1] H. Sanjoh, H. Yasaka, Y. Sakai, K. Sato, H. Ishii, and Y. Yoshikuni, "Multiwavelength light source with precise frequency spacing using a mode-locked semiconductor laser and an arrayed waveguide grating filter," *IEEE Photonics Technology Letters*, vol. 9, no. 6, pp. 818-820, 1997. DOI: [10.1109/68.585001](https://doi.org/10.1109/68.585001)
- [2] J. Ye, H. Schnatz, and L. W. Hollberg, "Optical frequency combs: from frequency metrology to optical phase control." *IEEE Journal of Selected Topics in Quantum Electronics*, vol. 9, no. 4, pp. 1041-1058, 2003. DOI: [10.1109/JSTQE.2003.819109](https://doi.org/10.1109/JSTQE.2003.819109)
- [3] M.-C. Lo, R. Guzmán, M. Ali, R. Santos, L. Augustin, and G. Carpintero, "1.8-THz-wide optical frequency comb emitted from monolithic passively mode-locked semiconductor quantum-well laser," *Optics Letters*, vol. 42, no. 19, pp. 3872-3875, 2017. DOI: [10.1364/OL.42.003872](https://doi.org/10.1364/OL.42.003872)
- [4] L. Hou, Y. Huang, Y. Liu, R. Zhang, J. Wang, B. Wang, H. Zhu, B. Hou, B. Qiu, and J. H. Marsh, "Frequency comb with 100 GHz spacing generated by an asymmetric MQW passively mode-locked laser," *Optics Letters*, vol. 45, no. 10, pp. 2760-2763, 2020. DOI: [10.1364/OL.392191](https://doi.org/10.1364/OL.392191)
- [5] D. Sun, D. Lu, R. Zhang, H. Wang, Q. Yang, and L. Zhao, "100-GHz ultra-short high-peak-power colliding-pulse mode-locked laser with asymmetric coating," *Optics Express*, vol. 31, no. 6, pp. 10533-10540, 2023. DOI: [10.1364/OE.483982](https://doi.org/10.1364/OE.483982)
- [6] D. Sun, Y. Niu, F. Guo, R. Zhang, L. Zhao, and D. Lu, "Generation of 10-GHz ultrashort pulse and flat optical comb using a semiconductor mode-locked laser," *Optics Communications*, vol. 583, 131655, 2025. DOI: [10.1016/j.optcom.2025.131655](https://doi.org/10.1016/j.optcom.2025.131655)
- [7] W. Shieh and K.-P. Ho, "Equalization-enhanced phase noise for coherent-detection systems using electronic digital signal processing," *Optics Express*, vol. 16, no. 20, pp. 15718-15727, 2008. DOI: [10.1364/OE.16.015718](https://doi.org/10.1364/OE.16.015718)
- [8] A. Kakkar, J. R. Navarro, R. Schatz, H. Louchet, X. Pang, O. Ozolins, G. Jacobsen, and S. Popov, "Comprehensive Study of Equalization-Enhanced Phase Noise in Coherent Optical Systems," *Journal of Lightwave Technology*, vol. 33, no. 23, pp. 4834-4841, 2015. DOI: [10.1109/JLT.2015.2491363](https://doi.org/10.1109/JLT.2015.2491363)
- [9] X. Fang, Y. Zhu, X. Cai, W. Hu, Z. He, S. Yu, and F. Zhang, "Overcoming laser phase noise for low-cost coherent optical communication," *Nature Communications*, vol. 15, 6339, 2024. DOI: [10.1038/s41467-024-50439-1](https://doi.org/10.1038/s41467-024-50439-1)
- [10] Y. Zhu, X. Fang, X. Cai, Y. Hu, W. Hu, and F. Zhang, "1780-km Reach Extension with Equalization-Enhanced Phase Noise Compensation Using 3-MHz LO for WDM Coherent Transmission," in *Proc. Asia Communications and Photonics Conference (ACP)*, 2024, paper 0801-157. DOI: [10.1109/ACP/IPOC63121.2024.10809786](https://doi.org/10.1109/ACP/IPOC63121.2024.10809786)
- [11] F. Buchali, A. Klekamp, L. Schmalen, and T. Drenski, "Implementation of 64QAM at 42.66 GBaud using 1.5 samples per symbol DAC and demonstration of up to 300 km fiber transmission," in *Proc. Optical Fiber Communication Conference (OFC)*, 2014, paper M2A.1. DOI: [10.1364/OFC.2014.M2A.1](https://doi.org/10.1364/OFC.2014.M2A.1)